

MAKING LARGE LANGUAGE MODELS BETTER REASONERS WITH ALIGNMENT

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ABSTRACT

Reasoning is a cognitive process of using evidence to reach a sound conclusion. The reasoning capability is essential for large language models (LLMs) to serve as the brain of the artificial general intelligence agent. Recent studies reveal that fine-tuning LLMs on data with the chain of thought (COT) reasoning process can significantly enhance their reasoning capabilities. However, we find that the fine-tuned LLMs suffer from an *Assessment Misalignment* problem, i.e., they frequently assign higher scores to subpar COTs, leading to potential limitations in their reasoning abilities. To address this problem, we introduce an *Alignment Fine-Tuning (AFT)* paradigm, which involves three steps: 1) fine-tuning LLMs with COT training data; 2) generating multiple COT responses for each question, and categorizing them into positive and negative ones based on whether they achieve the correct answer; 3) calibrating the scores of positive and negative responses given by LLMs with a novel constraint alignment loss. Specifically, the constraint alignment loss has two objectives: a) Alignment, which guarantees that positive scores surpass negative scores to encourage answers with high-quality COTs; b) Constraint, which keeps the negative scores confined to a reasonable range to prevent the model degradation. Beyond just the binary positive and negative feedback, the constraint alignment loss can be seamlessly adapted to the ranking situations when ranking feedback is accessible. Furthermore, we also delve deeply into recent ranking-based alignment methods, such as DPO, RRHF, and PRO, and discover that the constraint, which has been overlooked by these approaches, is also crucial for their performance. Extensive experiments on four reasoning benchmarks with both binary and ranking feedback demonstrate the effectiveness of AFT. In addition, AFT also performs well in multi-task and out-of-distribution situations.

1 INTRODUCTION

Reasoning is a cognitive process that involves utilizing evidence to reach a well-founded conclusion (Qiao et al., 2023; Huang & Chang, 2023). Recently, there has been a growing focus on enhancing the reasoning abilities of Large Language Models (LLMs) (Li et al., 2023b), particularly open-source LLMs (Yuan et al., 2023a; Luo et al., 2023; Mukherjee et al., 2023), because LLMs still lack reasoning skills (Wang et al., 2023b;d; Zheng et al., 2023) that are essential for them to serve as the brain of artificial general intelligence agents (Wang et al., 2023a; Yao et al., 2023; Song et al., 2023b).

Recent works (Chung et al., 2022; Hsieh et al., 2023; Mukherjee et al., 2023) find that training LLMs using data with a chain of thought (COT) reasoning process is a very effective method to improve the reasoning ability of LLMs. These studies typically train LLMs using maximum likelihood estimation (MLE), and employ a next-token prediction objective. However, MLE only assigns probability mass to the reference COT, which contradicts reasoning tasks where various reasoning paths can lead to the correct answer. In this paper, we find that previous vanilla fine-tuning (VFT) paradigm causes LLMs to suffer from an *Assessment Misalignment* problem, i.e., LLMs struggle with accessing the quality

Question: Weng earns \$12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?	Candidate Answer 1 (PPL 1.90): Weng earns $12/60 = <<12/60=0.2>>0.2$ \$ per minute Yesterday she did 50 minutes of babysitting, so she earned $50 * 0.2 = <<50*0.2=10>>10$ \$. ✓
Reference Answer (PPL 1.05): Weng earns $12/60 = \$<<12/60=0.2>>0.2$ per minute. Working 50 minutes, she earned $0.2 \times 50 = \$<<0.2*50=10>>10$.	Candidate Answer 2 (PPL 1.35): Weng earns \$12/hour $\times 60$ minutes = $\$<<12*60=720>>720$ per hour. Working 50 minutes, she earned $\$720 \times 50/60 = \$<<720*50/60=40>>40$. ✗

Figure 1: Perplexity of different answers given by the vanilla fine-tuning (VFT) LLM, where LLM assigns a lower perplexity to the incorrect candidate answer compared to the correct candidate answer.

of different COTs, ultimately limiting their reasoning capabilities. Take Figure 1 as an example, VFT-LLMs learn to generate the *Reference Answer* for the given *Question* by allocating probability mass to this *Reference Answer* and treating all other answers as negative outcomes. As a result, they struggle to assess the quality of other answers and tend to assign lower perplexity (higher score) to *incorrect Candidate Answer 1* compared to the *correct Candidate Answers 2*.

This behavior of VFT-LLMs is not consistent with that of humans, as humans have the ability to access the quality of different COTs after learning to reason. In addition, our pilot experiments (Section 3) find that after the same VFT process, the LLMs with better reasoning performance can give a more reasonable assessment to different COTs. Therefore, we hypothesize that we can improve the reasoning ability of LLMs by alleviating the assessment misalignment problem caused by VFT.

To address the assessment misalignment problem, in this paper, we propose an alignment fine-tuning (AFT) paradigm to improve LLM reasoning with three steps: **1)** fine-tuning LLMs using COT training data; **2)** generating multiple COT responses for each question using the fine-tuned LLMs, and categorizing them as positive and negative based on whether they deduce the correct answer; **3)** calibrating the scores of positive and negative responses given by LLMs with a novel constraint alignment (CA) loss. Specifically, the CA loss ensures that all positive scores (the scores of positive COTs) are larger than negative scores. In addition, the negative scores are protected by a constraint term, which is proven to be very important in preventing model degradation. Beyond just binary positive and negative feedback, the CA loss can be seamlessly adapted to ranking situations when ranking feedback is accessible. Furthermore, we also delve deeply into recent ranking-based methods for alignment, such as DPO (Rafailov et al., 2023), PRO (Song et al., 2023a) and RRHF (Yuan et al., 2023b), and find that the constraint, which has been overlooked by these approaches, is also crucial for their effectiveness.

In summary, our contributions are:

- 1)** We discover that LLMs fine-tuned by the vanilla fine-tuning (VFT) paradigm suffer from an Assessment Misalignment problem: they frequently assign lower scores to high-quality COTs compared to low-quality ones, which hinders their reasoning ability.
- 2)** We present an Alignment Fine-Tuning (AFT) paradigm, which comprises three straightforward steps with a novel constraint alignment loss to address the identified problem.
- 3)** We delve deeply into recent ranking-based methods for alignment and find that the constraint, which has been overlooked by these approaches, is also crucial for their performance.
- 4)** Experiments on four reasoning benchmarks with both binary and ranking feedback demonstrate the effectiveness of AFT. AFT also performs well in multi-task and out-of-distribution situations.

2 RELATED WORKS

2.1 IMPROVE REASONING OF LARGE LANGUAGE MODELS

Reasoning is a cognitive process that involves utilizing evidence to reach a well-founded conclusion, which is a core ability of LLMs to serve as the brain of the artificial general intelligence agent. Researchers have proposed a lot of methods to improve the reasoning ability of LLMs, which can be broadly divided into three groups: **1) pre-training:** The pre-training methods pre-train the LLMs on a vast of unsupervised datasets, such as the pile (Gao et al., 2020), the stack (Kocetkov et al., 2022), and so on, with a simple next token prediction objective. Researchers find that a larger model pre-trained on more data tends to have better reasoning ability (OpenAI, 2023; Anil et al.,

2023; Touvron et al., 2023); 2) *fine-tuning*: The fine-tuning methods can also enhance the reasoning ability of LLMs. Researchers have found that fine-tuning LLMs on the data with the reasoning chain-of-thought process can significantly improve the reasoning of LLMs (Mukherjee et al., 2023; Chung et al., 2022; Li et al., 2023a); 3) *prompting*: The prompting methods aims to improve the reasoning ability by carefully designed prompting strategy, such as chain-of-thought prompting (Wei et al., 2022), self-consistency (Wang et al., 2023c) strategy, and so on. The prompting methods do not change the model parameters, which is very convenient and practical. In this paper, we focus on the fine-tuning methods and find that traditional vanilla chain-of-thought fine-tuned LLMs suffer from an assessment misalignment problem, which hinders their reasoning ability. To this end, we propose an alignment fine-tuning paradigm to address this problem to enhance the reasoning ability of LLMs.

2.2 ALIGNMENT OF LARGE LANGUAGE MODELS

AI alignment research focuses on directing AI systems toward human-intended goals, preferences, or ethical principles. There are two primary categories of AI alignment methods: 1) *Reinforcement Learning from Human Feedback (RLHF)* (Ouyang et al., 2022), which trains a reward model by utilizing human feedback, which subsequently acts as a reward function for optimizing an agent’s policy through reinforcement learning (RL) techniques, such as Proximal Policy Optimization (Schulman et al., 2017). RLHF is employed to align powerful LLMs, like ChatGPT and GPT-4. However, RL-based methods face limitations concerning training efficiency and complexity; 2) *Supervised Fine-tuning with Ranking* (Liu et al., 2022; Yuan et al., 2023b; Song et al., 2023a; Rafailov et al., 2023), which involves training LLMs using a supervised fine-tuning paradigm and incorporating a ranking loss to help LLMs align with human preferences. Previous alignment research has mainly focused on improving the safety of LLMs, frequently neglecting the importance of alignment for reasoning. Furthermore, widely used ranking methods often neglect the constraint term when reducing scores of low-quality examples, which can potentially have a negative impact on model performance. In this paper, we point out the effectiveness of alignment for reasoning and introduce a novel constraint alignment loss to make LLMs better reasoners with alignment.

3 PILOT EXPERIMENTS

In this section, we first briefly introduce the vanilla fine-tuning (VFT) paradigm, and then we demonstrate the assessment misalignment problem of VFT for reasoning.

3.1 VANILLA FINE-TUNING

VFT finetunes LLMs on a dataset $\{(q_i, c_i, a_i)\}_{i=1}^N$ with N examples. Each example consists of a question q_i , a COT reasoning process c_i , and an answer a_i . The LLMs are finetuned to generate the reference response $r_i = [c_i; a_i]$ based on q_i with a MLE objective loss function:

$$\mathcal{L}_{VFT} = - \sum_{j=1}^{|r_i|} \log P(r_{i,j} \mid r_{i,<j}, q_i; \theta). \quad (1)$$

where θ is the model parameter and $r_{i,j}$ is the j -th token of r_i .

3.2 ASSESSMENT MISALIGNMENT OF VFT FOR REASONING

Intuitively, the MLE objective seeks to exclusively allocate probability mass to the reference COT c_i for question q_i , which does not correspond with the characteristics of reasoning tasks, where the correct COT is not limited to the reference one. This objective uniformly treats all other correct and incorrect COTs as negative examples. As a result, it will impede LLMs from learning to assess the quality of various COTs and degrade their reasoning ability.

To demonstrate this, we first fine-tune LLama-7B, LLama-13B, LLama2-7B, and LLama2-13B on the training data of GSM8k and ECQA with Equation 1 (please refer to Section 5.1 for the detailed VFT settings). Then, for each question q_i in the training data, we use VFT-LLMs to generate three positive COTs $\{c_i^{p1}, c_i^{p2}, c_i^{p3}\}$ that induce to the correct answer and three negative COTs $\{c_i^{n1}, c_i^{n2}, c_i^{n3}\}$ that